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Supplementary Information for

The translatome of neuronal cell bodies, dendrites and axons.

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Datasets S1 to S9

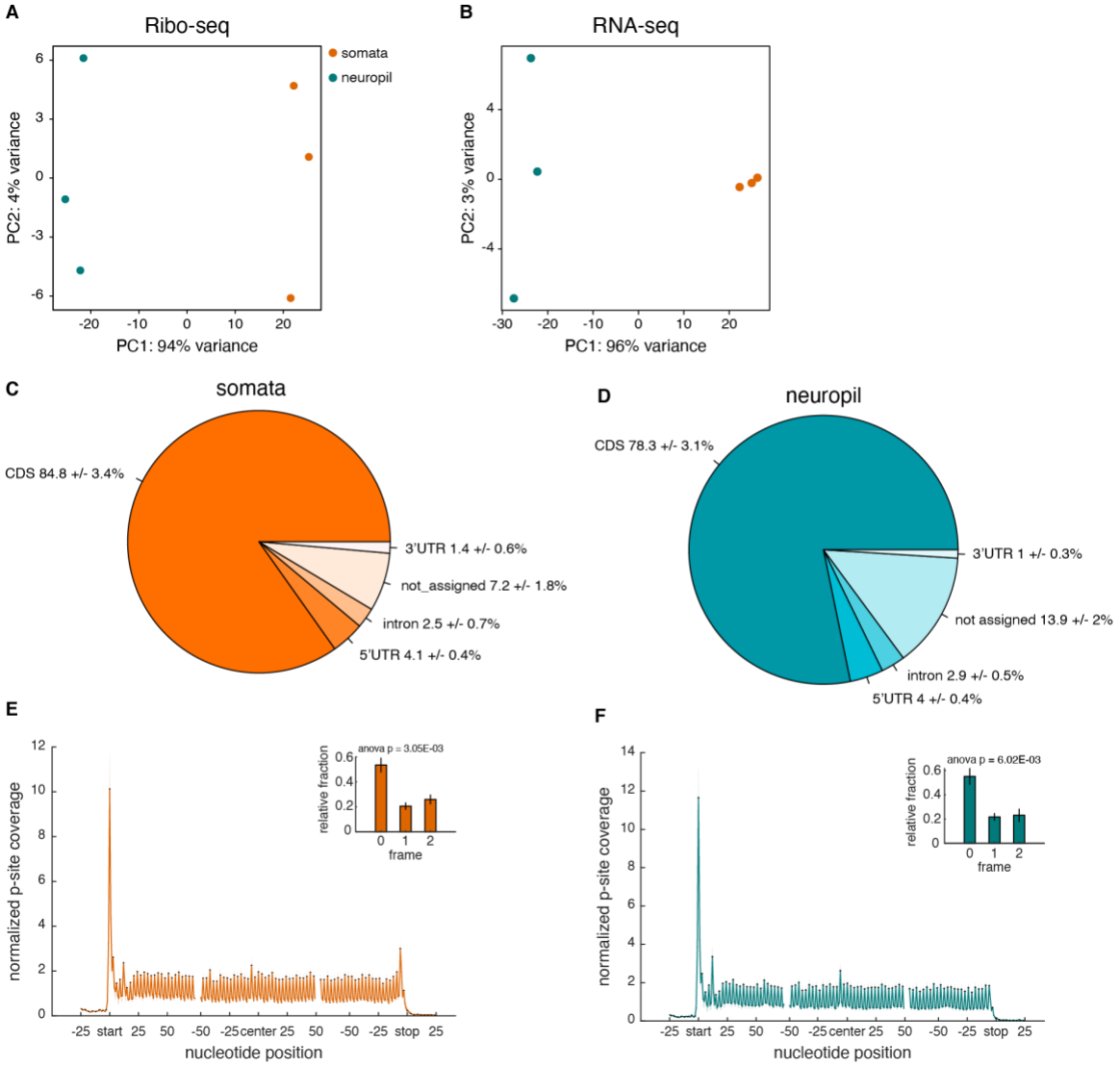

Fig. S1. Quality metrics of the RNA- and Ribo-seq data. (A-B) Principal component analysis (PCA) showing the within and between group variance of neuropil and somata Ribo-seq (A) and RNA-seq (B) samples. (C-D) Percentage of ribosomal footprints of the somata (C) and neuropil (D) libraries, mapping to different genomic regions, including the coding sequence (CDS), 5'-/3' untranslated regions (UTR) and introns. (E-F) Metagene analyses showing the P-site coverage of ribosomal footprints in the somata (E) and neuropil (F) samples. The mean normalized coverage is projected per nucleotide position around the start codon (start), central portion (center) and stop codon (stop) of the CDS. The standard error is shaded (n = 3). Bars in the insets indicate standard errors. The inset shows the relative fraction of the footprints in different reading frames. p = 3.05e-03 (E) and p = 6.02e-03 (F), One-way ANOVA.

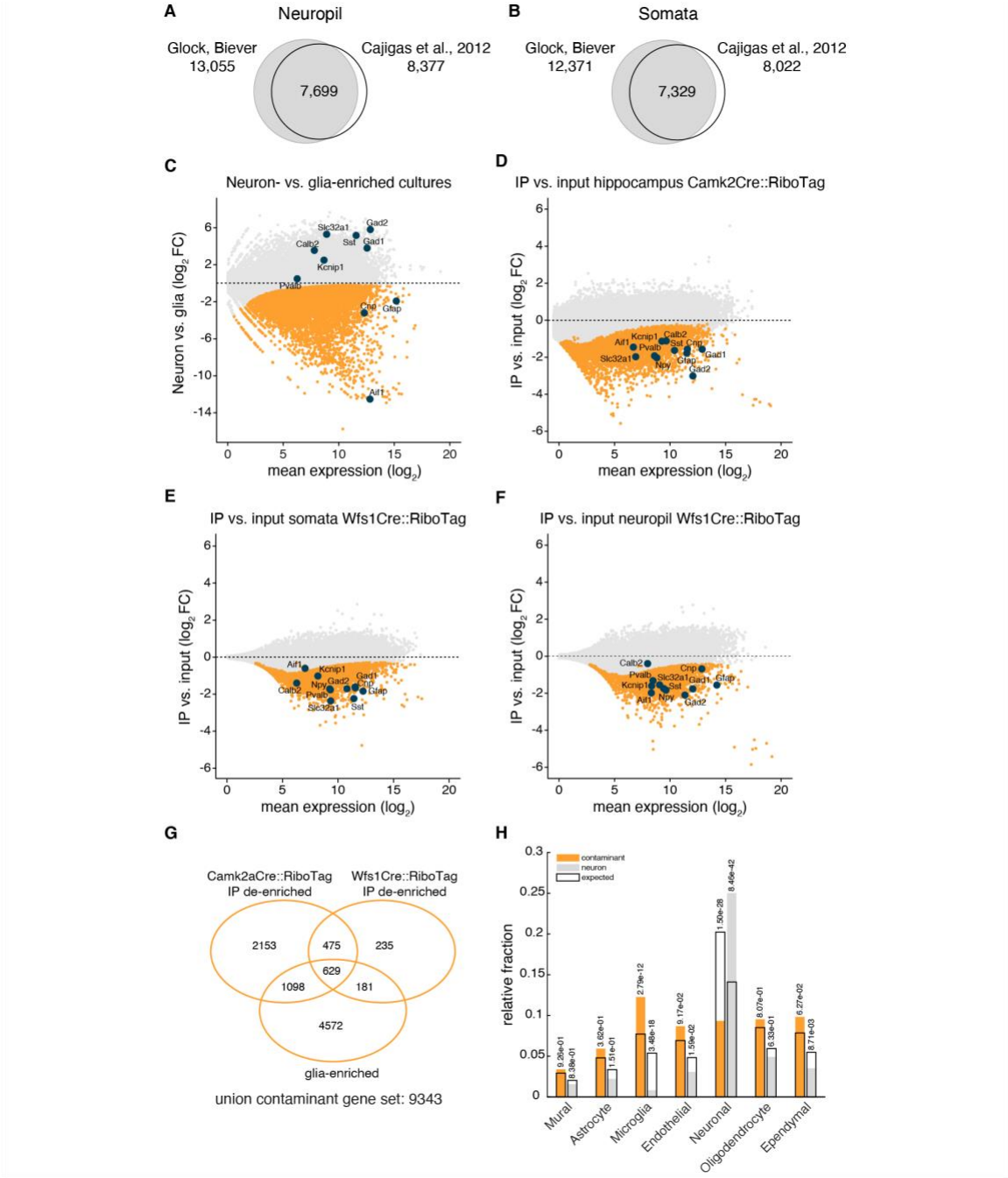

Fig. S2. Bioinformatic filtering for neuronal transcripts. (A-B) Venn diagrams showing the overlap between transcripts with 1 CPM in two out of three neuropil (A) or somata (B) Ribo-seq replicates and previously published transcriptome data from microdissected layers (1). (C) Identification of transcripts enriched in glia. MA plot representing differentially expressed transcripts in neuron- vs. glia-enriched cultures. Differential expression analysis revealed 6480 transcripts significantly enriched in glia (yellow). DESeq2, FDR < 0.01. The blue dots highlight key non-excitatory neuron marker genes. Note the positive foldchanges of interneuron markers, highlighting the presence of this cell type in neuron-enriched cultures. (D, E and F) Identification of transcripts expressed in hippocampal glia and interneurons. MA plot representing the transcripts differentially expressed between the Rpl22-HA immunoprecipitation (IP) and input from hippocampi of Camk2a-Cre::RiboTag (D), the microdissected somata (E) and neuropil (F) from Wfs1-Cre::RiboTag mice.

Transcripts significantly de-enriched in the Rpl22-HA IP are highlighted in yellow. Blue dots represent key interneuron and glia markers. DESeq2, FDR < 0.05. (G) Venn diagram comparing the genes significantly de-enriched in the Rpl22-HA IP from the hippocampi of Camk2a-Cre- (D), the microdissected somata and neuropil layers of Wfs1-Cre-RiboTag mice (E and F) and glia-enriched genes (C). The union of these datasets represents the list of “contaminant” genes expressed in glia and interneurons. Transcripts with an experimentally-validated dendritic localization were rescued from the filter (1). (E) Observed-to-expected ratio of expressed contaminant (yellow) or excitatory neuron (gray) genes defined as described in (G) and associated with different hippocampal cell types by a previous study (2).

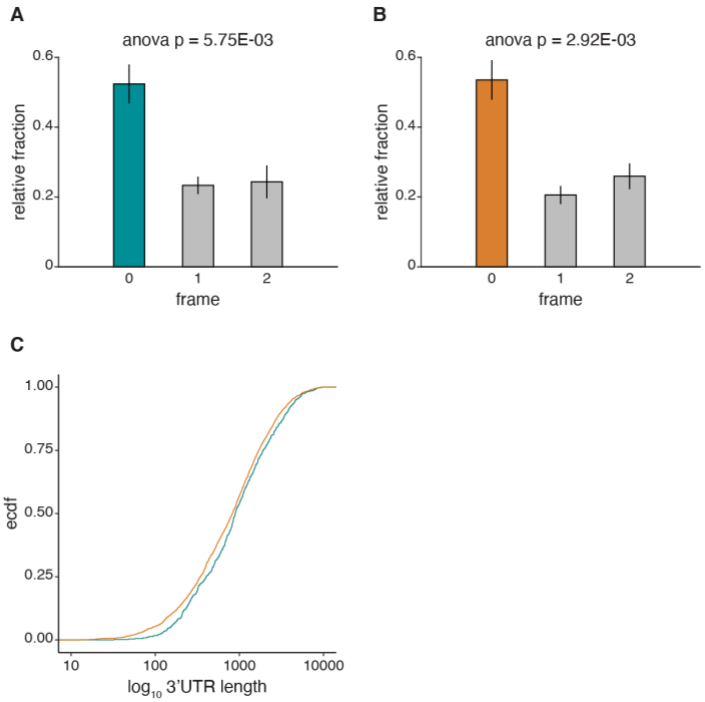

Fig. S3. Three nucleotide periodicity of neuropil- and somata-translation-up transcripts. Relative fraction of the footprints in different reading frames for neuropil-translation-up transcripts in the neuropil (A) and somata-translation-up transcripts in the somata (B). p = 5.75e-03 (A) and p = 2.92e-03 (B), One-way ANOVA. (C) Empirical cumulative distribution function (ecdf) plot of 3'UTR lengths (log_{10}) for neuropil-translation-up (teal) and somata-translation-up (orange) genes. p = 0.04152, Kolmogorov-Smirnov test.

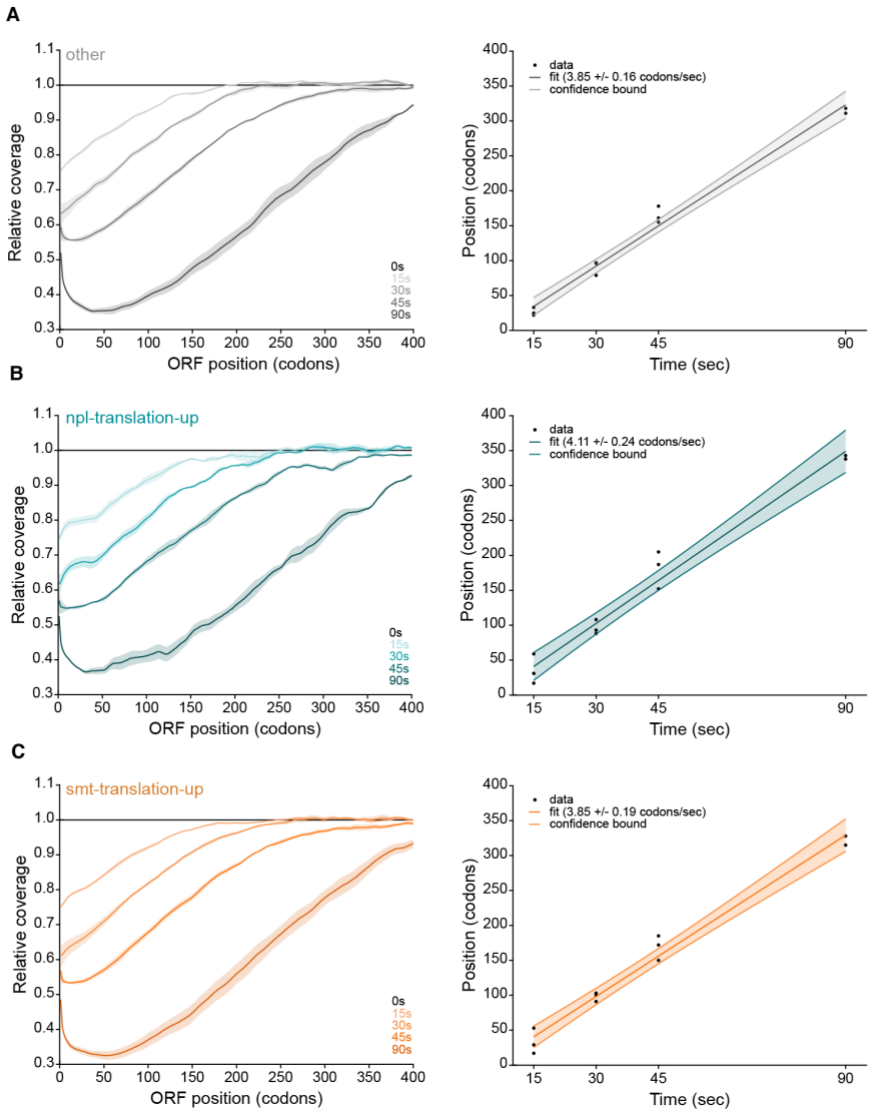

Fig. S4. Measuring elongation rates of neuropil- and somata-translation-up transcripts. Progressive loss of ribosome footprints from the 5' end of other (gray, A), neuropil (npl)-translation-up (teal, B) and somata (smt)-translation-up (orange, C) transcripts in rat hippocampal cultures treated with harringtonine. Ribosome densities of the samples treated with harringtonine for 15, 30, 45 and 90 s were normalized to the untreated control (n = 3). The codon position of 15% ribosome depletion was plotted as a function of harringtonine incubation times for smt-translation-up (orange), npl-translation-up (teal) and other (gray) transcripts. A linear fit was computed for each transcript subset. The standard errors are shaded. Elongation rates for smt-translation-up (orange), npl-translation-up (teal) and other (gray) transcripts inferred from the slope of the linear fit.

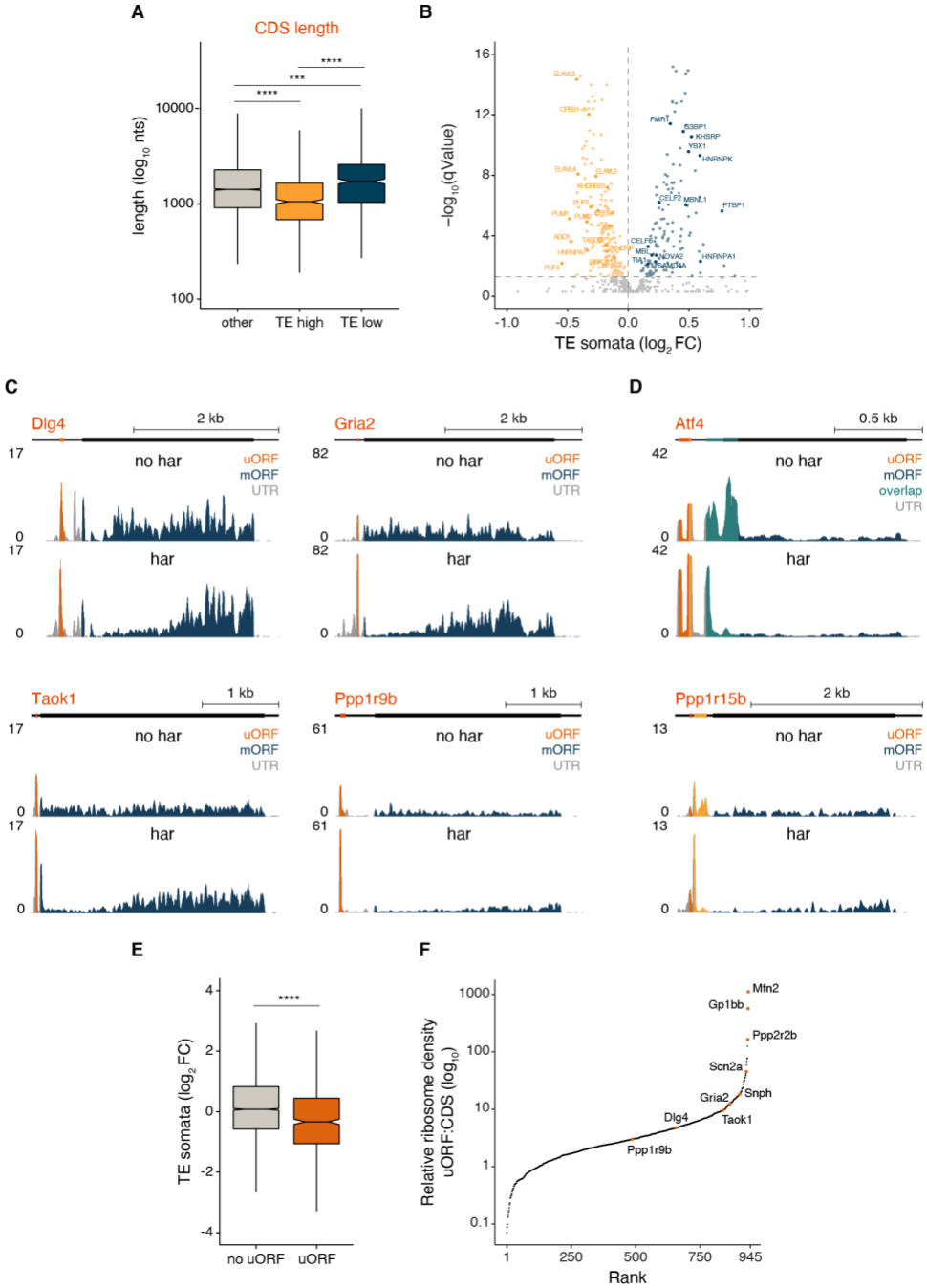

Fig. S5. Features of translationally regulated transcripts in the neuropil and somata. (A) Box plots of coding sequence (CDS) length ($\log_{10}$ nucleotides (nts)) for TE_{high} (yellow), TE_{low} (blue) and other (gray) genes. Bars indicate 1.5*IQR. *** $p < 0.001$; **** $p < 0.0001$; one-way ANOVA test followed by pairwise t-test with BH p-value adjustment. (B) Shown are RNA-binding protein (RBP) motifs within 3'UTRs associated with significantly lower (blue) or higher (yellow) somata TE values (qValues < 0.05; Wilcoxon rank sum test; see Methods). (C-D) Coverage tracks representing the average ribosome footprint reads along UTRs (gray), uORFs (orange), overlapping uORFs (teal) or the main protein coding sequence (blue) in untreated (no har) or harringtonine-treated (har) primary hippocampal neuron cultures. The harringtonine-induced ribosome footprint accumulation at upstream start sites enabled the detection of novel uORFs within *Dlg4*, *Gria2*, *Taok1* and *Ppp1r9b* (C) and confirmed the well-studied uORF usage within the

transcripts *Atf4* and *Ppp1r15b* (D). The y-axis indicates reads per million (RPM). (E) Somata TE (log_2FC) measurements of transcripts containing translated uORFs ("uORF") or not ("no uORF"). **** $p < 0.0001$; Welch two-sample t-test. (F) Rank plot showing neuropil translated uORFs sorted by their relative uORF:CDS ribosome density (log_{10}).

Sample_name	Biosample_id	Bioproject_id
Neuropil_Total (TaxID: 10116)	SAMN12135214	PRJNA550323
Somata_Total (TaxID: 10116)	SAMN15014696	PRJNA550323
Neuropil_RNA (TaxID: 10116)	SAMN12135219	PRJNA550323
Somata_RNA (TaxID: 10116)	SAMN15014697	PRJNA550323
Wfs1Cre_RiboTag_neuropil_IP (TaxID: 10090)	SAMN12135220	PRJNA550323
Wfs1Cre_RiboTag_neuropil_Input (TaxID: 10090)	SAMN12135221	PRJNA550323
Wfs1Cre_RiboTag_somata_IP (TaxID: 10090)	SAMN12135222	PRJNA550323
Wfs1Cre_RiboTag_somata_Input (TaxID: 10090)	SAMN12135223	PRJNA550323
Camk2Cre_RiboTag_hippocampus_IP (TaxID: 10090)	SAMN12135224	PRJNA550323
Camk2Cre_RiboTag_hippocampus_Input (TaxID: 10090)	SAMN12135225	PRJNA550323
Hippocampal_Neurons_control_0s (TaxID: 10116)	SAMN15014698	PRJNA550323
Hippocampal_Neurons_harringtonine_15s (TaxID: 10116)	SAMN15014699	PRJNA634994
Hippocampal_Neurons_harringtonine_30s (TaxID: 10116)	SAMN15014700	PRJNA550323
Hippocampal_Neurons_harringtonine_45s (TaxID: 10116)	SAMN15014701	PRJNA634994
Hippocampal_Neurons_harringtonine_90s (TaxID: 10116)	SAMN15014702	PRJNA550323
Hippocampal_Neurons_harringtonine_150s (TaxID: 10116)	SAMN15014703	PRJNA634994
Neuron_enriched (TaxID: 10116)	SAMN15014704	PRJNA634994
Glia_enriched (TaxID: 10116)	SAMN15014705	PRJNA634994

Table S1. Availability of raw high-throughput datasets.

Dataset S1 (separate file). Differential RNA-seq analysis between neuron- and glia-enriched cultures.

Dataset S2 (separate file). Differential Ribo-seq analysis between neuronal compartments.

Dataset S3 (separate file). Differential RNA-seq analysis between neuronal compartments.

Dataset S4 (separate file). Translational efficiency values of neuronal transcripts in the somata, neuropil and comparison between the two compartments.

Dataset S5 (separate file). Neuropil median translational efficiency values of neuronal transcripts sharing the same 3'UTR RBP motifs.

Dataset S6 (separate file). Somata median translational efficiency values of neuronal transcripts sharing the same 3'UTR RBP motifs.

Dataset S7 (separate file). Neuropil median translational efficiency values of neuronal transcripts sharing the same 5'UTR RBP motifs.

Dataset S8 (separate file). Somata median translational efficiency values of neuronal transcripts sharing the same 5'UTR RBP motifs.

Dataset S9 (separate file). Positional information about translated uORFs in the neuropil and somata.

References

1. I. J. Cajigas, et al., The local transcriptome in the synaptic neuropil revealed by deep sequencing and high-resolution imaging. Neuron 74, 453–466 (2012).
2. A. Zeisel, et al., Brain structure. Cell types in the mouse cortex and hippocampus revealed by single-cell RNA-seq. Science 347, 1138–1142 (2015).